

Introduction to Electronics Engineering (IEE)



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IEE Time Table

Time Days	9:30 - 10:30	10:30 - 11:30	11:30 - 12:30	12:30 - 1:30	1:30 - 2:30	2:30 - 3:30	3:30 - 4:30	4:30 - 5:30
Mon				IEE (BS 304)	Lunch Break	IEE LAB		
Tue	IEE (EB 302)	IEE LAB						
Wed								
Thu			IEE (BS 202)					
Fri						IEE (BS 303)		
Sat								

IEE Theory

Course Code	EC 101
Teaching Hours/ Week	(3-0-1-0)
Credits	03
Total Contact Hours	42

Course Objective

1	To understand PN Junction, diodes and their applications. .
2	To comprehend BJT, FET and their bias configurations.
3	To analyze feedback in amplifier circuits, op amp and its applications.
4	To analyze different number system, Logic Gates and applications of Boolean algebra.
5	To understand fundamentals of communication networks & its applications

Syllabus

Module 1 (10hrs.)

Semiconductor Diodes and their Applications: Introduction to Electronics Engineering, PN junction diodes; Characteristics and Calculation of diode-resistance, temperature dependence, diode-capacitance; Zener Diode – Operation and Applications; Diode as a Rectifier, Diode Clipper and Clamper

Module 2 (10hrs.)

Bipolar Junction Transistors (BJT): PNP and NPN Transistors, Construction and Characteristics of CB, CE and CC Configurations, dc and ac load line analysis, operating point, Transistor biasing: Fixed bias, emitter bias/self-bias, Junction Field Effect Transistors: Construction, Characteristics, Operations

Syllabus

Module 3 (08hrs.)

Sinusoidal Oscillators: Concept of positive and negative feedback, Barkhausen criterion, Determination of Frequency and Condition of oscillation, RC Phase Shift and Wein bridge oscillator. Operational Amplifiers: Characteristics, Applications of Inverting and non-inverting amplifiers.

Module 4 (08hrs.)

Introduction to Digital Electronics: Boolean Algebra, Symbolic representation, Boolean algebraic function, Realization of Basic logic gates using universal gates, Adder, Subtractor.

Module 5 (08hrs.)

Introduction to electronic communication system: Electromagnetic Communication spectrum band and applications, Elements of Electronic Communication System, analog vs. digital communication, Need for modulation; Introduction to IoT, Evolution of 1G to 5G and beyond.

Referred Study Materials

Text Books:

1. Electronic Circuits Analysis and Design, Third Edition, Donald A Neamen, The McGraw-Hill Companies
2. MICROELECTRONIC CIRCUITS, Seventh Edition, A. S. Sedra and K.C. Smith, Oxford University Press, India, 2017
3. Millman J., Halkias C.C., Parikh Chetan, "Integrated Electronics: Analog and Digital Circuits and Systems", Tata McGraw-Hill, 2/e.
4. Mano M.M., "Digital Logic and Computer Design", Pearson Education, Inc, Thirteenth Impression, 2011.
5. Singal T. L., "Analog and Digital Communications", Tata McGraw-Hill, 2/e.
6. Haykin S., Moher M., "Introduction to Analog & Digital Communications", Wiley India Pvt. Ltd., 2/e.

Reference Books:

1. Robert L. Boylestad and Louis Nashelsky, "Electronic Devices and Circuit Theory", Pearson Education, 10th Edition, 2009
2. Malvino and Brown, "Digital Computer Electronics," Tata McGraw - Hill Education, 2001.
3. R. A. Gayakwad, "Op-Amps and Linear Integrated Circuits," PHI Learning, 2009.
4. T.L. Floyd, "DIGITAL FUNDAMENTALS", Pearson Education, 11th Edition.

Open Sources:

1. Prof. Mahesh B. Patil, "Basic Electronics", NPTEL – IIT Bombay.
<https://nptel.ac.in/courses/108/101/108101091/>



COURSE OUTCOME

Outcome	At the end of the course, the learner will be able to	Bloom's Level	Expected proficiency percentage	Expected Attainment percentage
CO-1	Apply the concept of working principle of diode(s) in the Rectifiers, Clippers, Clampers, Voltage Regulator, and LED circuits	3	70%	65%
CO-2	Analyse simple amplifier circuits using the concepts of BJT, JFET, and OP-AMP and their applications.	4	70%	60%
CO-3	Design simple logic circuits by using the concept of number systems, logic gates, numeric codes, and Boolean algebra minimization	6	70%	70%
CO-4	Understand the fundamentals of communication technology & its applications	2	70%	60%

Chapter 1

Introduction

What is Electronics ?

- Electronics is the branch of science and engineering which deals with the study of flow of electrons through vacuum, gas, or semiconductor.
- An electronic circuit is a closed path formed by interconnection of electronic components through which electric current flows to perform a specific function

Types of Electronics:

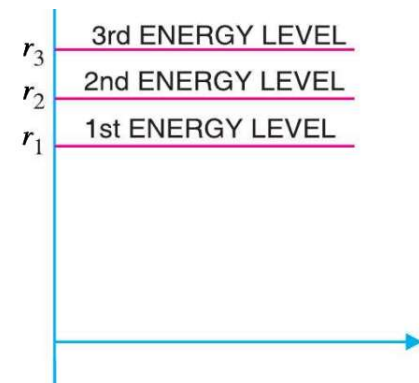
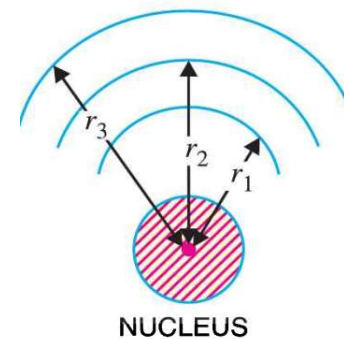
- Analog Electronics
- Digital Electronics
- Power Electronics
- Microelectronics
- Optoelectronics
- Communication Electronics

Atomic Structure

- An atom is the smallest particle of an element that retains the chemical properties of that element.
- Atomic structure refers to the structure of atom comprising of a nucleus (center) in which the protons (positively charged) and neutrons (neutral) are present.
- The negatively charged particles called **electrons revolve around the center of the nucleus**.
- The electrons in each permitted orbit have a certain fixed amount of energy. The larger the orbit (i.e. larger radius), the greater is the energy of electrons.
- If an electron is given additional energy (e.g. heat, light etc.), it is lifted to the higher orbit. The atom is said to be in a state of excitation.

Atomic weight = no of protons + no of neutrons.

Atomic number = no of protons = no of electrons in atom



Semiconductor

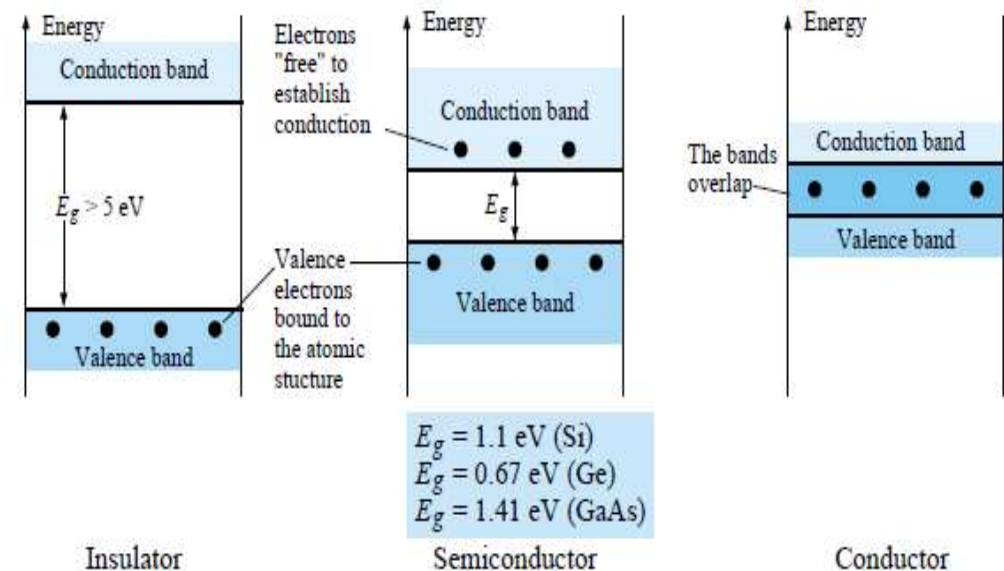
- In terms of their electrical properties, materials can be classified into three groups: conductors, semiconductors, and insulators.
- A **semiconductor** is a material whose **electrical conductivity lies between that of a conductor and an insulator** and whose conductivity can be **controlled** by adding impurities (doping) or applying external fields.
- The behavior of electrons in solids is governed by their arrangement in energy bands.

Conduction Band: The energy band where electrons are free to move and contribute to electrical conductivity.

Valence Band: The energy band where electrons are bound to atoms and do not contribute to conductivity.

Fermi Level: The highest energy level that an electron can occupy at the absolute zero temperature is known as the Fermi Level.

The Fermi level lies between the valence band and conduction band

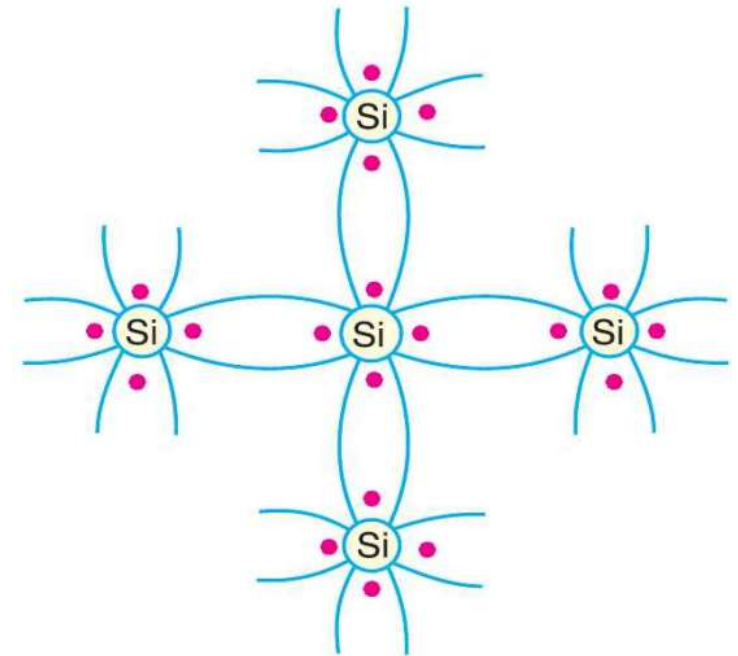
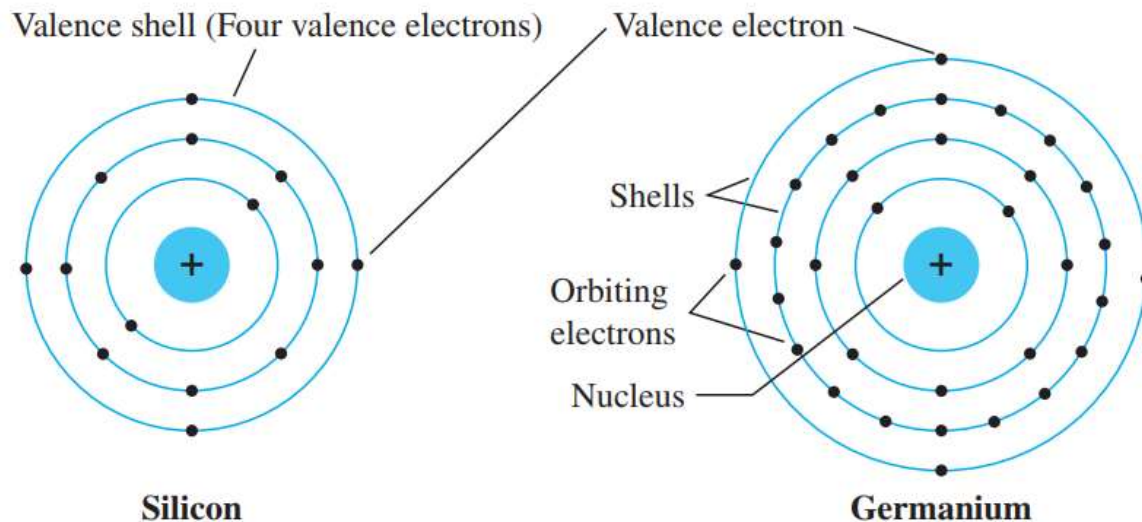


Comparison Table of Materials

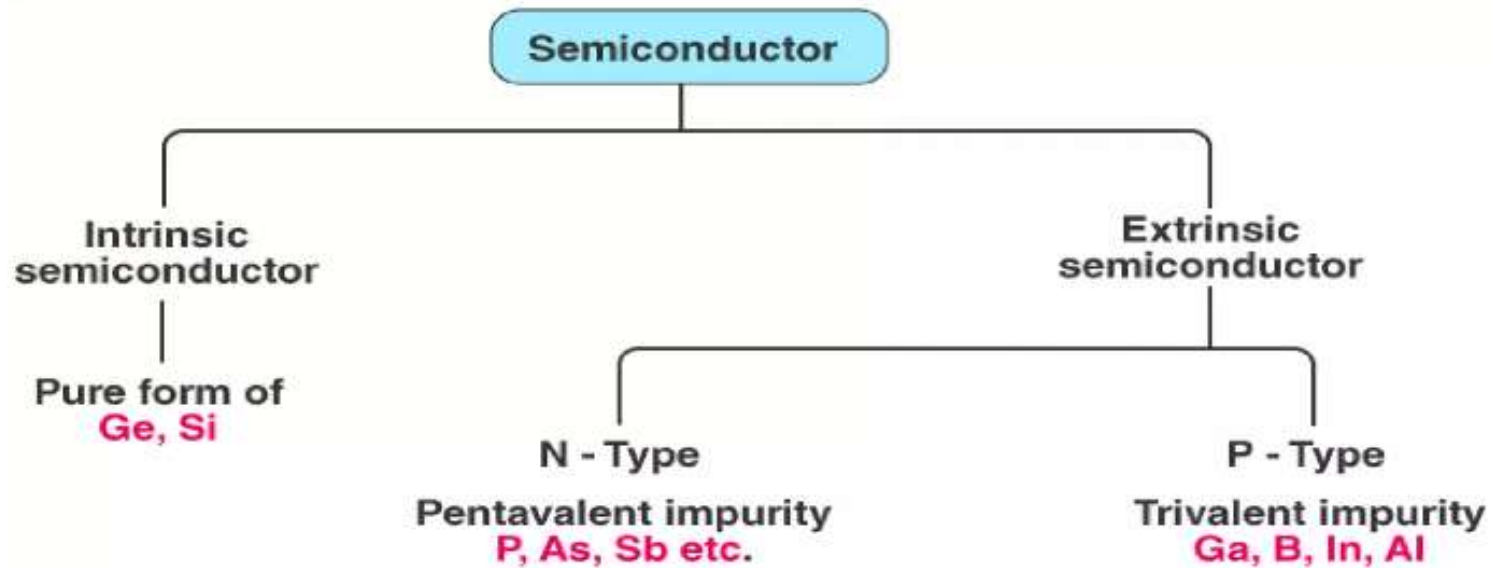
Parameter	Conductor	Semiconductor	Insulator
Forbidden Energy Gap	Not Exist	Small (1eV)	Large (>5eV)
Resistivity	Low	Moderate	High
Flow of current	Due to movement of free electrons.	Due to movement of electrons and holes.	Almost negligible but only due to free electrons.
Charge carriers in conduction band	Completely filled	Partially filled	Completely vacant
Charge carriers in valence band	Almost vacant	Partially filled	Completely filled
Example	Copper, Aluminium, graphite etc.	Silicon, Germanium, arsenic etc.	Paper, rubber, glass, plastic etc.
Applications	Conducting wires, Transformers, in electrical cords etc.	Diodes, transistors, opto couplers etc.	Sports equipment, home appliances etc.

Semiconductor

- In general, semiconductor materials fall into one of two classes: *single-crystal* and *compound*.
- The three semiconductors used most frequently in the construction of electronic devices are ***Ge, Si, and GaAs***.
- In a pure silicon or germanium crystal, the four valence electrons of one atom form a bonding arrangement with four adjoining atoms ***is called covalent bonding***.



Types of Semiconductors



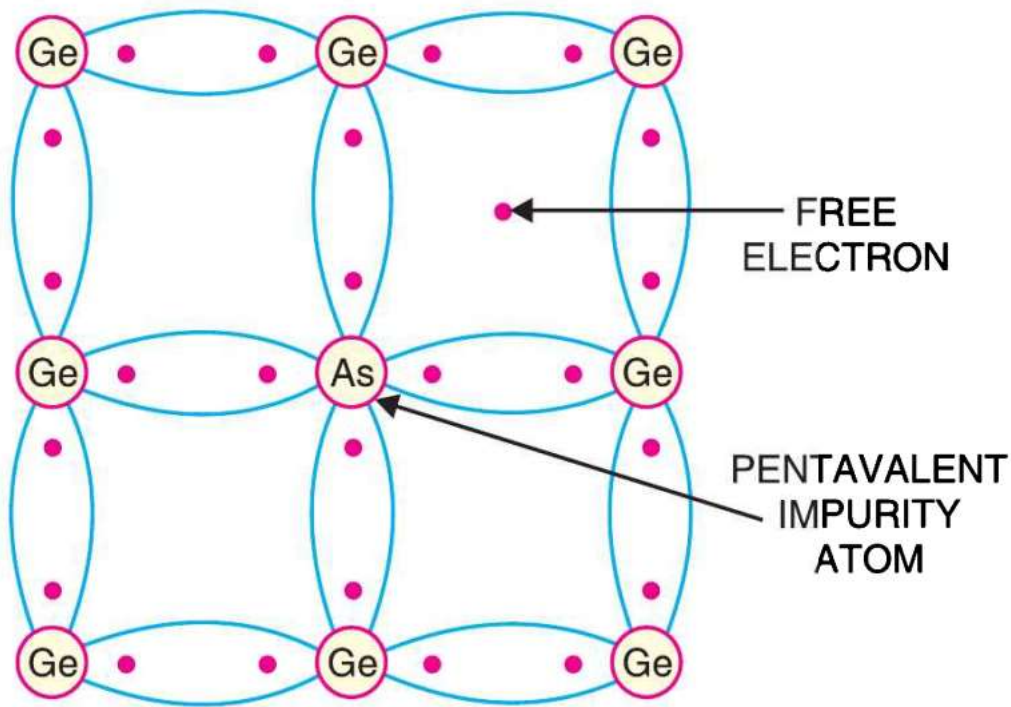
Intrinsic semiconductors

- Intrinsic semiconductors are pure form semiconductor materials without any significant impurities. Examples Silicon (Si) and germanium (Ge).
- In an intrinsic semiconductor, the number of electrons in the conduction band (where they can move freely) equals the number of holes (empty states in the valence band where an electron could exist).

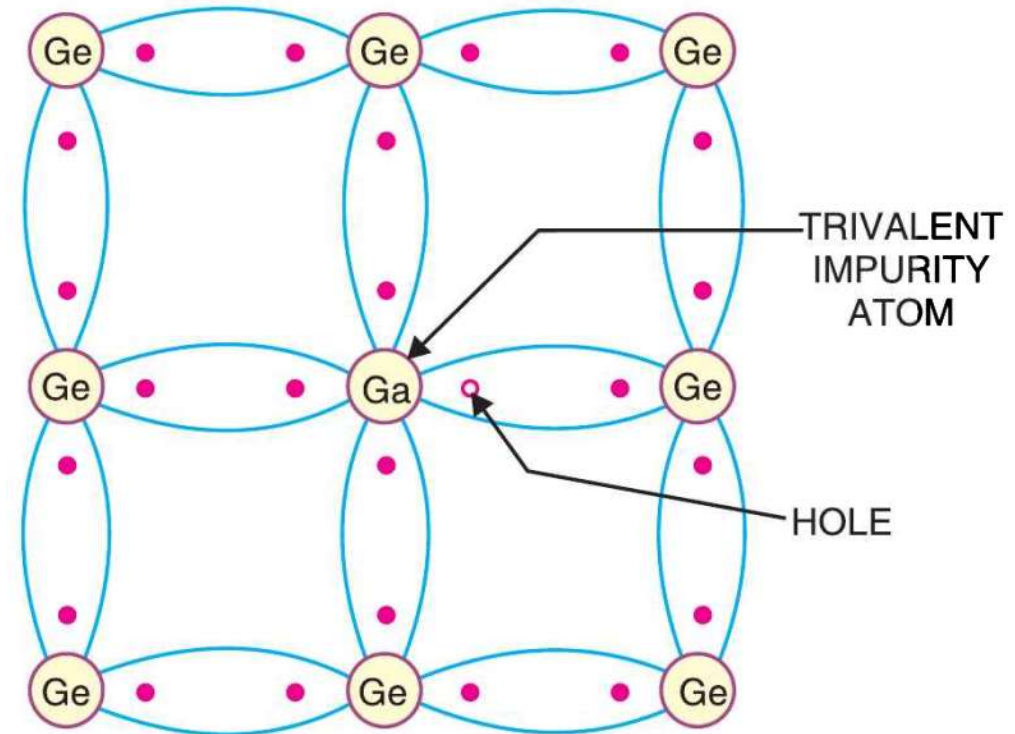
Extrinsic Semiconductors

- Extrinsic semiconductors are created by adding specific impurities, called dopants, to an intrinsic semiconductor to enhance its conductivity.
- Types of Extrinsic Semiconductors:
 - ✓ **n-type:** An *n*-type material is created by adding impurity elements that have *five* valence electrons (***pentavalent***), such as *antimony*, *arsenic*, and *phosphorus*. *The diffused impurities with five valence electrons are called donor atoms.*
 - ✓ **p-type:** An *p*-type material is created by adding impurity elements that have *three* valence electrons (***trivalent***), such as *boron*, *gallium*, and *indium*. *The diffused impurities with three valence electrons are called acceptor atoms.*
- The controlled addition of dopants allows precise control over the electrical properties of semiconductors, making them essential in devices like diodes, transistors, and integrated circuits.

Extrinsic Semiconductors



Arsenic impurity in n-type material



Gallium impurity in p-type material

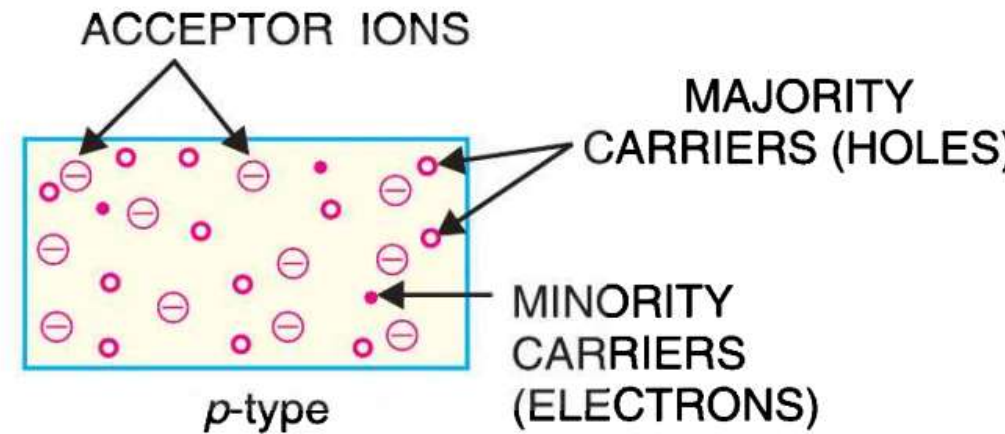
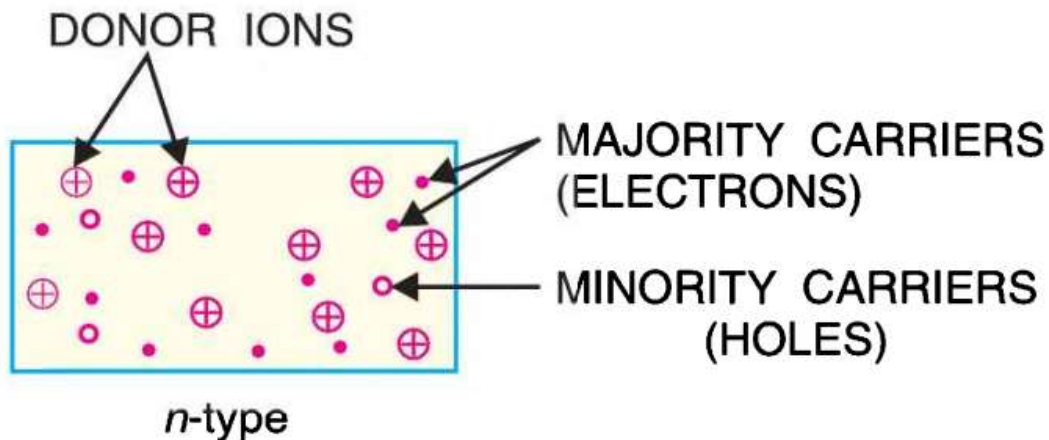
Intrinsic Semiconductors Vs Extrinsic Semiconductors

Property	Intrinsic Semiconductor	Extrinsic Semiconductor (n-type & p-type)
Purity	Pure (no impurities)	Doped with impurities
Conductivity	Low at room temperature	Higher due to the presence of free charge carriers
Charge Carriers	Equal number of electrons and holes	Majority and minority carriers (electrons or holes)
Effect of Temperature	Conductivity increases with temperature	Conductivity depends on doping and temperature
Types	Not applicable	n-type (electron majority) and p-type (hole majority)
Energy Band Structure	No additional energy levels	Donor or acceptor energy levels present

Majority and Minority Carriers

*In an **n-type** material the **electron** is called the **majority** carrier and the **hole** the **minority** carrier.*

*In a **p-type** material the **hole** is the **majority** carrier and the **electron** is the **minority** carrier*



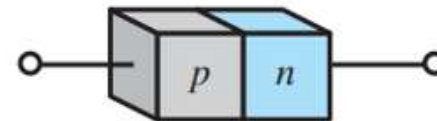
PN Junction Diode

- The **PN junction** is the fundamental building block of semiconductor devices such as diodes, transistors, and integrated circuits.
- The PN junction diode is created by simply joining an n -type and a p -type material together.
- **A diode has three operating conditions:**

No bias

Forward bias

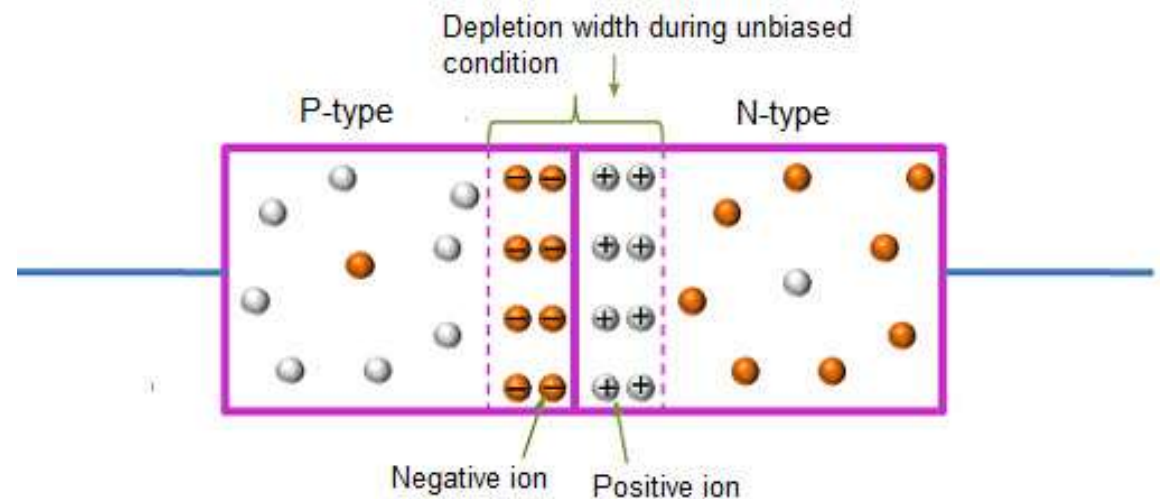
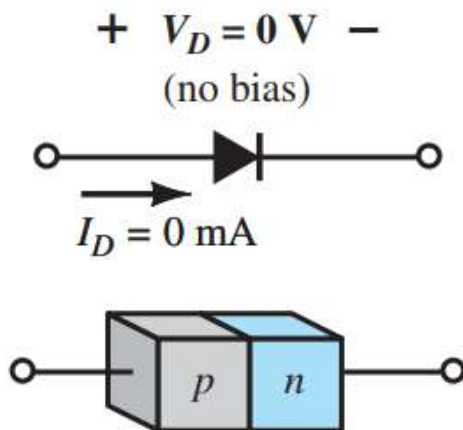
Reverse bias



- The term *bias* refers to the application of an external voltage across the two terminals of the device to extract a response.

PN Junction :No Bias

- Electrons from n-region diffuse into p-region. Holes from p-region diffuse into n-region. Movement continues until balance is established.
- Electrons and holes recombine near junction, creating a region with no free charge carriers. Only immobile ions remain (negative ions on p-side, positive ions on n-side).
- An electric field develops across the depletion region. The flow of electrons and holes due to diffusion is balanced by the force of this electric field.
- In the absence of an applied bias across a semiconductor diode, the net flow of charge in one direction is zero.



Characteristics of the Depletion Region

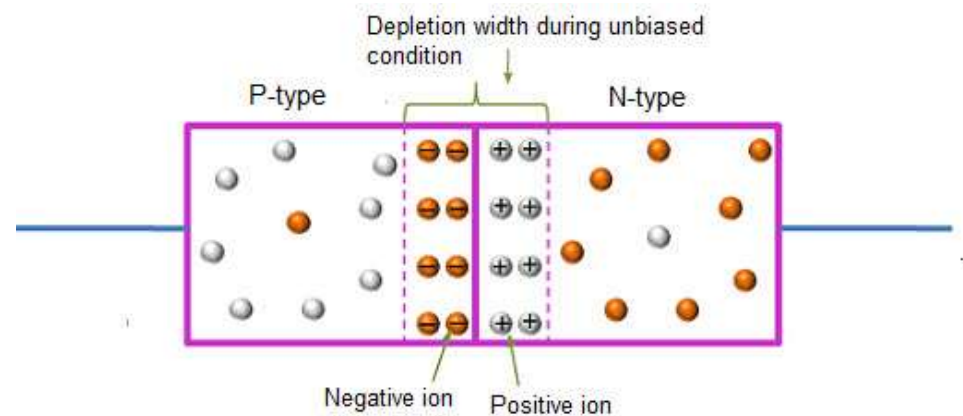
- **Width:** Depends on doping concentration. Heavily doped regions have narrower depletion regions.
- **Charge Distribution:** p-region contains negatively charged acceptor ions; n-region contains positively charged donor ions.
- **Electric Field:** Points from positively charged n-region to negatively charged p-region.

Barrier Potential

The electric field creates a potential barrier preventing further diffusion of charge carriers.

Built-in potential (V_b) is typically:

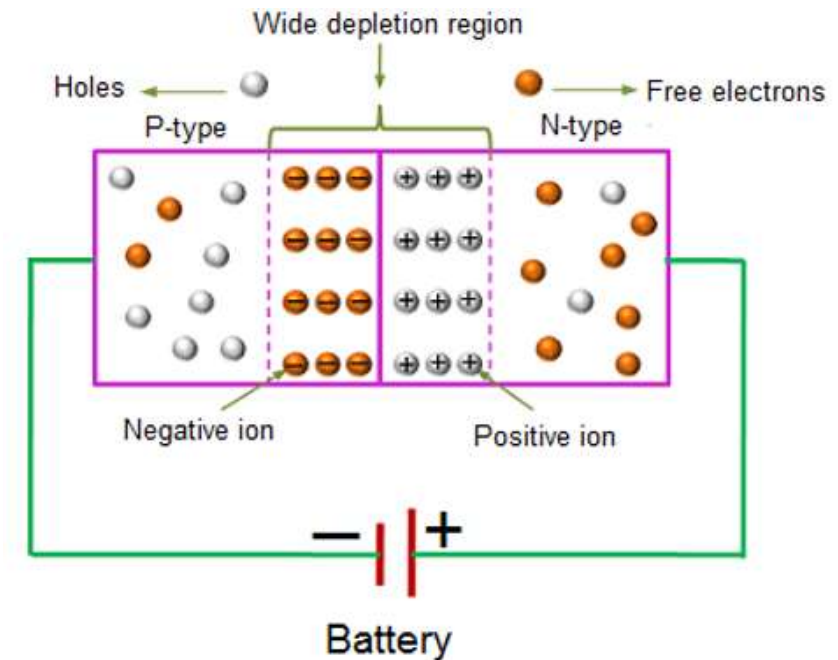
- 0.7V for silicon
- 0.3V for germanium



Biasing of PN Junction: Reverse Bias

When positive voltage is applied to n-region and negative to p-region:

- Potential barrier is **increased**
- Depletion region widens
- Charge carriers cannot cross the junction
- Very little current flows (minority charge carrier)
- If reverse voltage exceeds critical value, breakdown can occur through Zener or avalanche breakdown

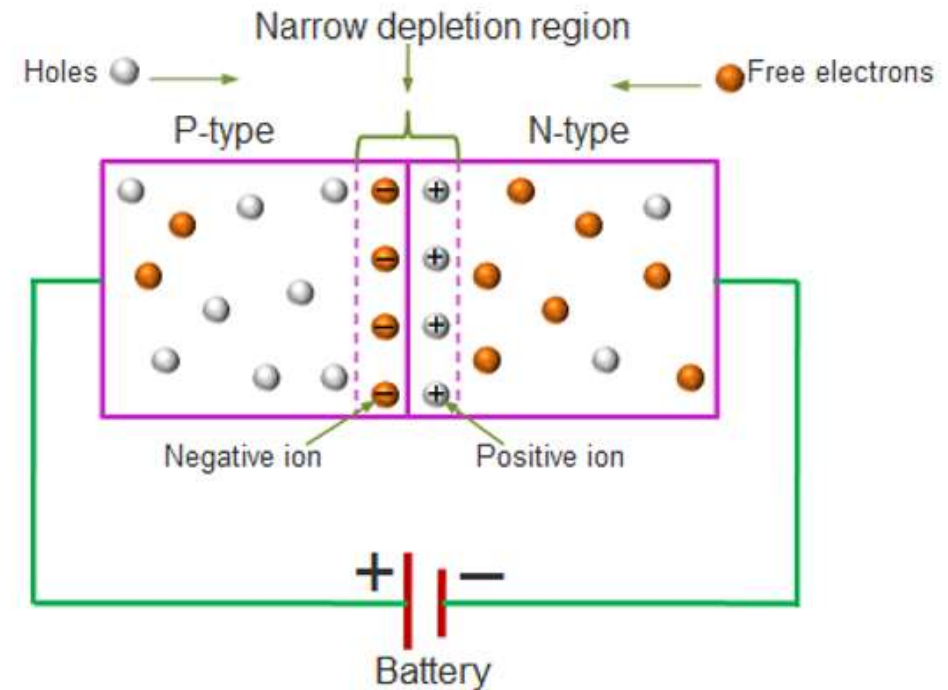


- Reverse saturation current (I_s) or leakage current is caused by thermally generated minority carriers swept across the junction by the built-in electric field.
- It is a very small, almost constant with reverse voltage and strongly increases with temperature.

Biasing of PN Junction: Forward Bias

When positive voltage is applied to p-region and negative to n-region

- Potential barrier is **reduced**
- Charge carriers can cross the junction
- Electrons move toward p-region, holes toward n-region
- Results in **forward current** flow
- In silicon diodes, forward voltage must be $\sim 0.7\text{V}$ for significant current.



Biasing of PN Junction: Forward Bias

General characteristics of a semiconductor diode : Shockley's equation

$$I_D = I_s(e^{V_D/nV_T} - 1) \quad (\text{A})$$

I_s is the reverse saturation current

V_D is the applied forward-bias voltage across the diode

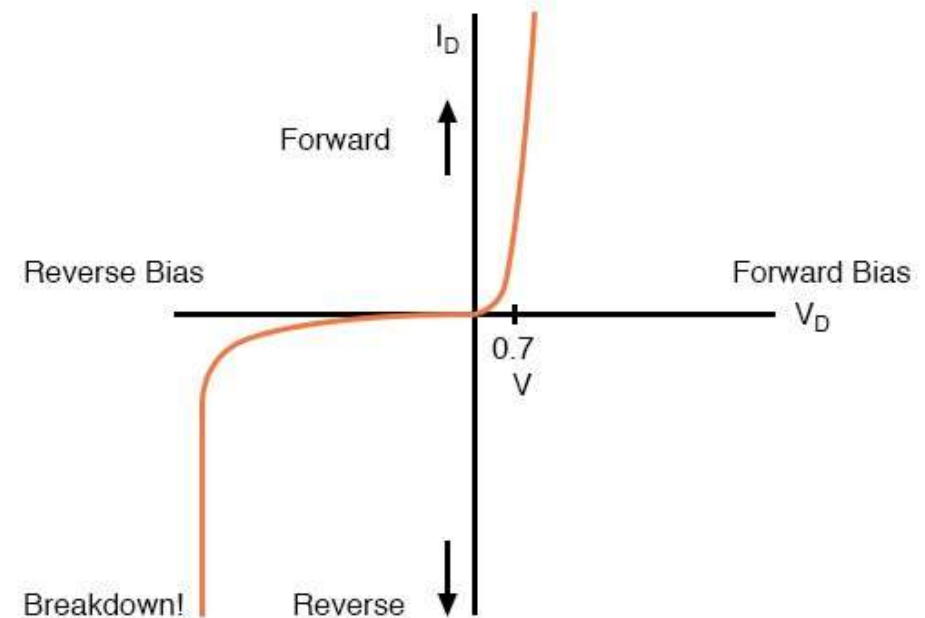
n is the ideality factor, and $n=1$ for Germanium and $n=2$ for Silicon

$$V_T = \frac{kT_K}{q} \quad (\text{V})$$

k is Boltzmann's constant = 1.38×10^{-23} J/K

T_K is the absolute temperature in kelvins = $273 +$ the temperature in $^{\circ}\text{C}$

q is the magnitude of electronic charge = 1.6×10^{-19} C



Current-Voltage (I-V) Characteristics

Forward Bias

- When voltage exceeds threshold (0.7V for silicon), current increases exponentially
- Diode conducts current easily

Reverse Bias

- Only very small reverse saturation current flows
- If reverse voltage exceeds breakdown voltage, diode conducts large currents (may damage regular diodes)

Knee Voltage: It is the minimum voltage after which the forward current starts increasing exponentially is called knee voltage

